

Ehsan Al Twal

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SKILLS

Electrical: Electrical Systems Design, PCB Design (Altium), BLDC Motor Control, Microcontrollers (STM32, Teensy 4.0, Arduino), FPGAs (De1-SoC), Debugging & Soldering, SPI, I2C, CAN

Software: NN, CNN, SIFT, ROS, CV, Java, C, C++, C#, OOP, Python, Assembly, VHDL, Game Dev, MATLAB

Mechanical: Designing, Prototyping, Testing, 3D printing, CAD (Onshape, Solidworks)

EDUCATION

4th Year Engineering Physics Undergrad, University of British Columbia

Sep 2021 – Jun 2026

- Specialized in Robotics and Electrical Engineering

Vancouver, Canada

PROFESSIONAL EXPERIENCE

Electrical Engineer, Sarcomere Dynamics

May 2024 – Present

- Engineered electrical system and firmware for a new robotic hand, **3 times cheaper** and designed for flammable environments, enabling new market entry.
- Redesigned another robotic hand's electrical system with BLDC motors. Designed motor driver and controller PCBs in Altium Designer, and developed a field-oriented control library, extending product lifespan from **1 to 5 years**.
- Enhanced motor control accuracy through current sensing and a magnetic sensor, replacing Hall effect sensors, resulting in a **more compact and cheaper product**.

Vancouver, Canada

Mechanical/Materials Engineer, Matthews International - Terrella Energy Systems

Jan 2023 – Apr 2023

- Led a project to improve material strength and adhesion by **40%** through mechanical testing and behavioural analysis under varying conditions
- Developed innovative mechanical testing methods, identified necessary equipment, conducted experiments, and presented findings to enhance material performance.

Vancouver, Canada

PROJECTS

UBC Formula Electric, Electrical Engineer - Altium Designer, PCB Design, Laboratory Proficiency, LTSpice

- Designed the BMS Monolithic Module PCB using Altium to balance and monitor battery voltage and temperature, covering PCB design, assembly, component selection, and LTSpice simulation.
- Developed a Li-ion battery simulation setup enabling debugging of six BMS modules without holding up the power supply queue, accelerating team testing processes.
- Performed online verification and validation of various PCB schematics through Jira

Robot Race Car Competition, Electrical Engineer - STM32, C++, Circuit Design & Debugging

- Engineered an H-Bridge circuit for motor control and integrated key electronic circuits, including a rotary encoder, servo motor, and regulators. Achieved fastest lap in competition by **5 seconds**
- Fused sensor arrays (magnetometer, IMU, accelerometer, gyroscope) for accurate 3D orientation and utilized laser distance and ultrasound sensors for precise localization.
- Devised a track mapping approach via Desmos, programming the robot for autonomous route navigation

Autonomous ROS Robot with CNN-Based Text Recognition,

Python, CNN, Computer Vision, ROS, PID, Linux

- Programmed a vehicle to navigate autonomously while obeying traffic laws in a ROS environment.
- Implemented OpenCV for real-time detection of pedestrians, cars, road elements, and roadside posters.
- Developed a convolutional neural network for text recognition, achieving **100% accuracy** in reading images of roadside posters.